

STA 2101: Statistics & Probability

Lecture 05: Charts and Graphs Continued

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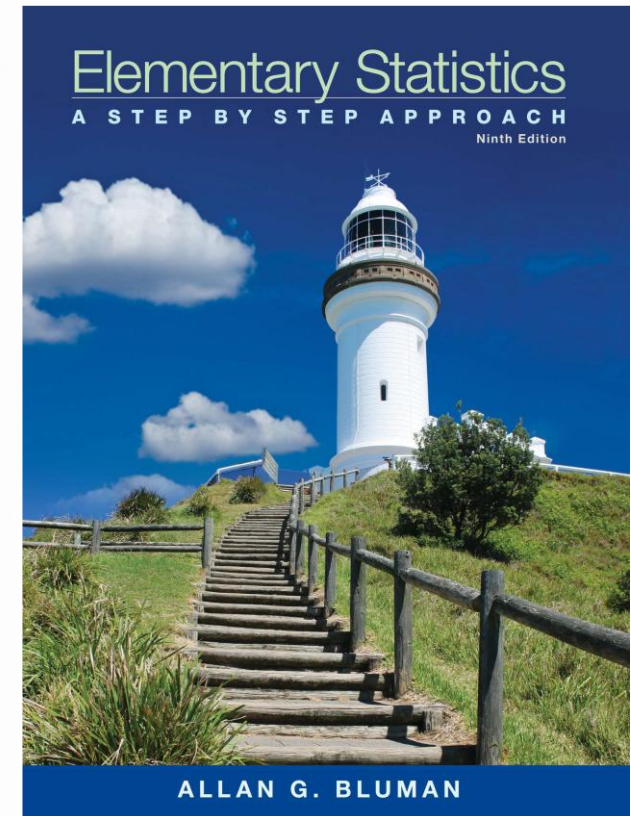
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Class Code
Sec 1 - es2uztok
Sec 5 - ayvzsvgm

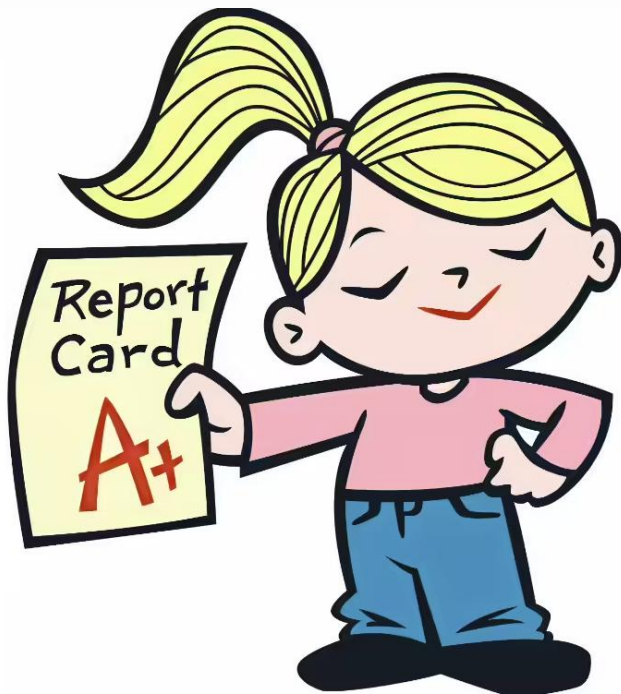
Spring 2026

Reference Book

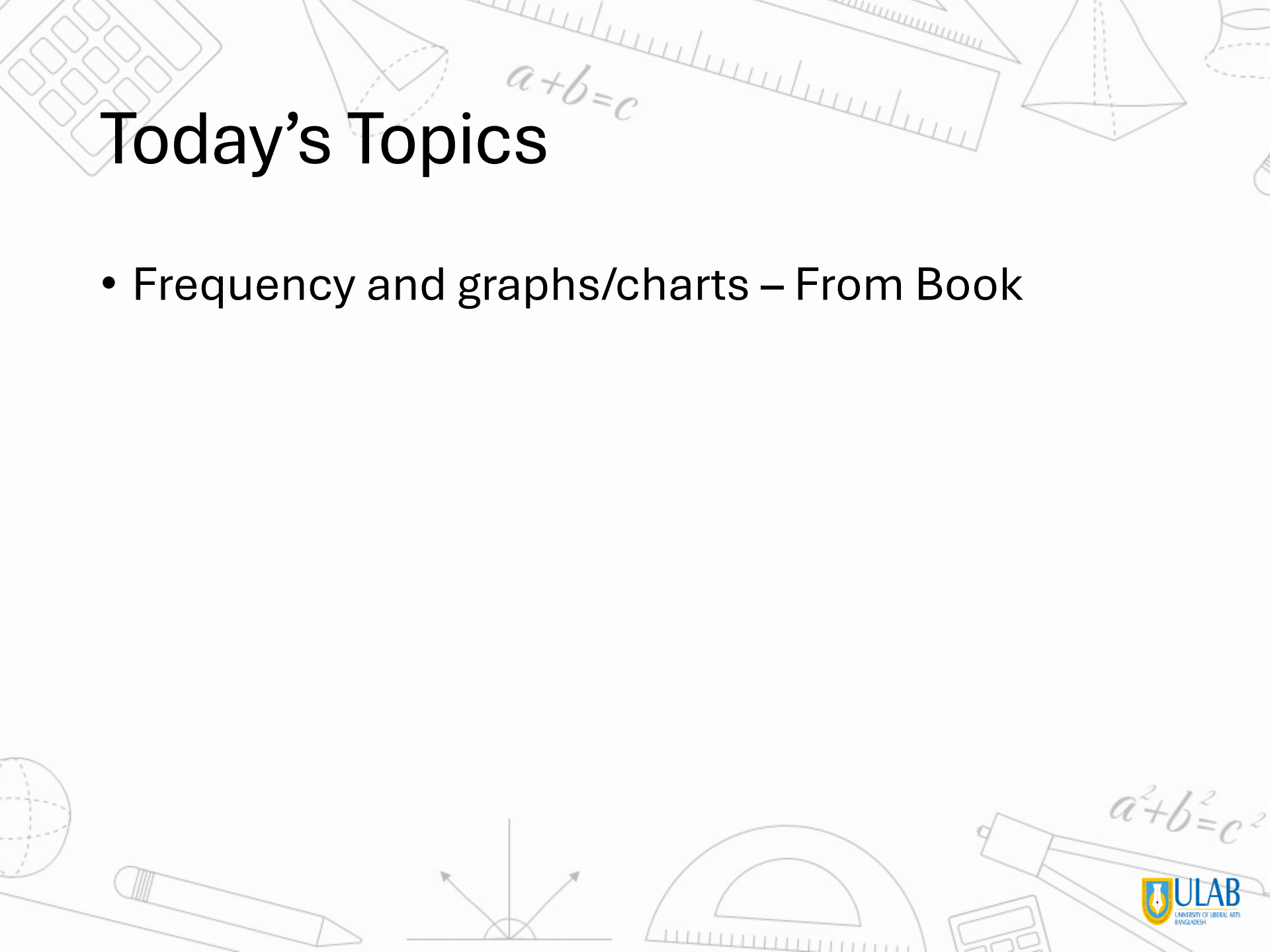
- **Text Book(s):**
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver
- **Reference Material/Book(s):**
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



Mark Distribution



Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	5
Project	25
Total	100

The background of the slide is decorated with various mathematical and geometric illustrations. At the top, there is a ruler, a pencil, and a cone. In the center, the equation $a+b=c$ is written in a cursive font. To the right, there is a pyramid and a cylinder. At the bottom, there is a globe, a pencil, a protractor, a compass, and the equation $a^2+b^2=c^2$. The University of Liberal Arts Bangladesh (ULAB) logo is located in the bottom right corner.

Today's Topics

- Frequency and graphs/charts – From Book

Relative Frequency Graphs

The histogram, the frequency polygon, and the ogive shown previously were constructed by using frequencies in terms of the raw data. These distributions can be converted to distributions using *proportions* instead of raw data as frequencies. These types of graphs are called **relative frequency graphs**.

EXAMPLE 2-7 Miles Run per Week

Construct a histogram, frequency polygon, and ogive using relative frequencies for the distribution (shown here) of the miles that 20 randomly selected runners ran during a given week.

Class boundaries	Frequency
5.5–10.5	1
10.5–15.5	2
15.5–20.5	3
20.5–25.5	5
25.5–30.5	4
30.5–35.5	3
35.5–40.5	2
	<u>20</u>
	Total

Practice
Now

SOLUTION

Step 1 Convert each frequency to a proportion or relative frequency by dividing the frequency for each class by the total number of observations.

For class 5.5–10.5, the relative frequency is $\frac{1}{20} = 0.05$; for class 10.5–15.5, the relative frequency is $\frac{2}{20} = 0.10$; for class 15.5–20.5, the relative frequency is $\frac{3}{20} = 0.15$; and so on.

Place these values in the column labeled Relative frequency.

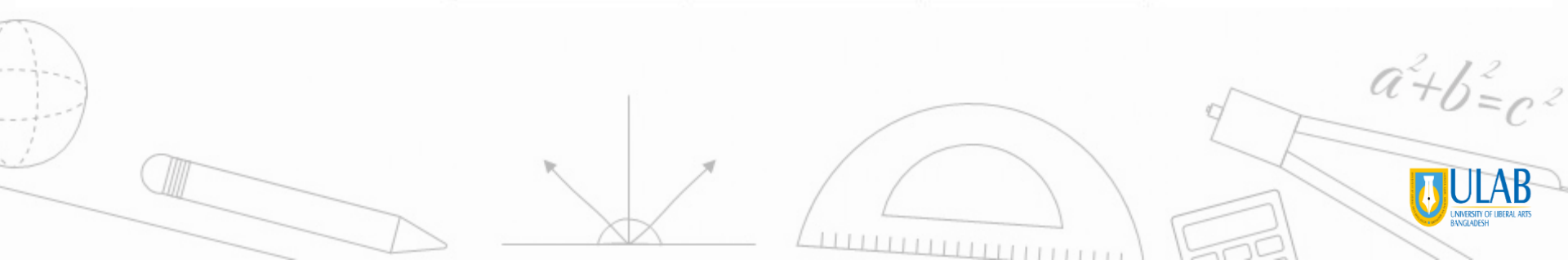
Class boundaries	Midpoints	Relative frequency
5.5–10.5	8	0.05
10.5–15.5	13	0.10
15.5–20.5	18	0.15
20.5–25.5	23	0.25
25.5–30.5	28	0.20
30.5–35.5	33	0.15
35.5–40.5	38	0.10
		<u>Total 1.00</u>


$$a+b=c$$

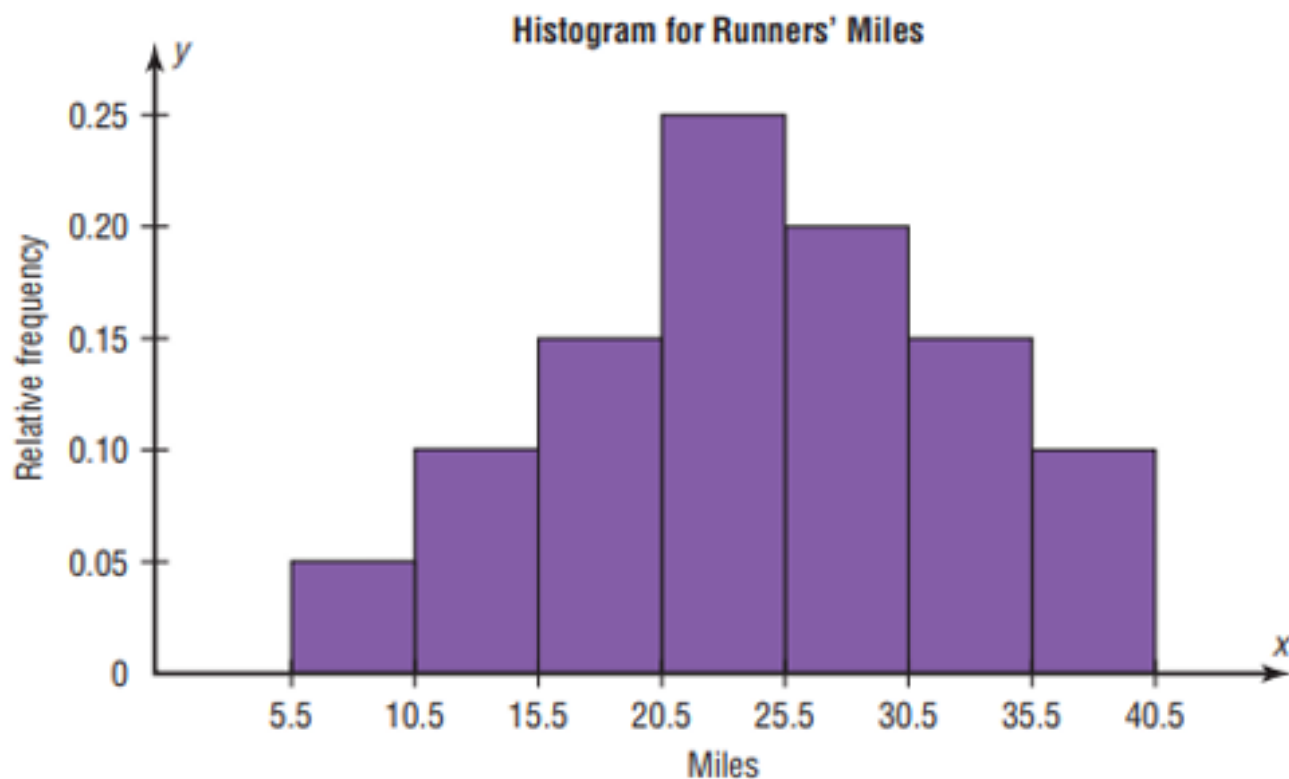
Step 2 Find the cumulative relative frequencies. To do this, add the frequency in each class to the total frequency of the preceding class. In this case, $0 + 0.05 = 0.05$, $0.05 + 0.10 = 0.15$, $0.15 + 0.15 = 0.30$, $0.30 + 0.25 = 0.55$, etc. Place these values in the column labeled Cumulative relative frequency.

An alternative method would be to find the cumulative frequencies and then convert each one to a relative frequency.

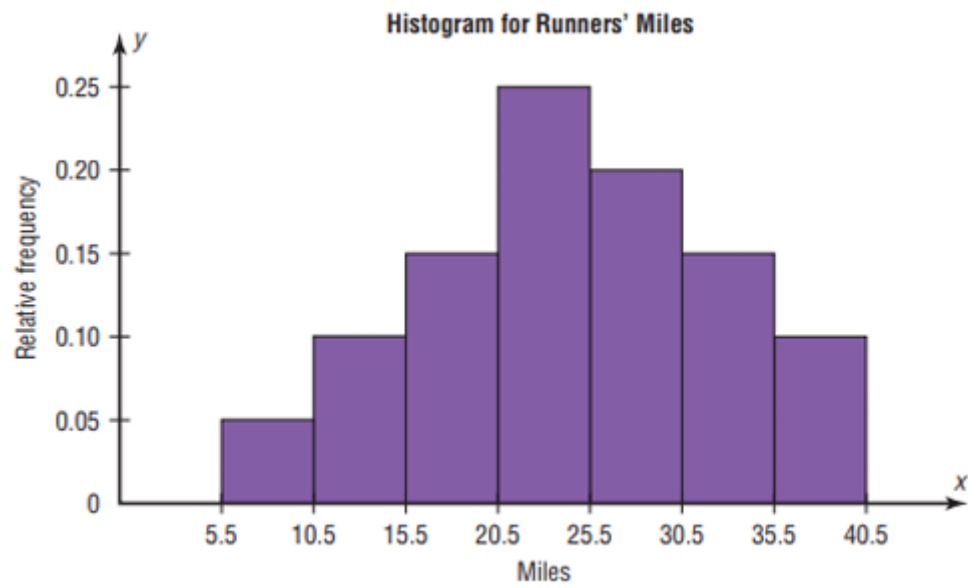
	Cumulative frequency	Cumulative relative frequency
Less than 5.5	0	0.00
Less than 10.5	1	0.05
Less than 15.5	3	0.15
Less than 20.5	6	0.30
Less than 25.5	11	0.55
Less than 30.5	15	0.75
Less than 35.5	18	0.90
Less than 40.5	20	1.00


$$a^2+b^2=c^2$$

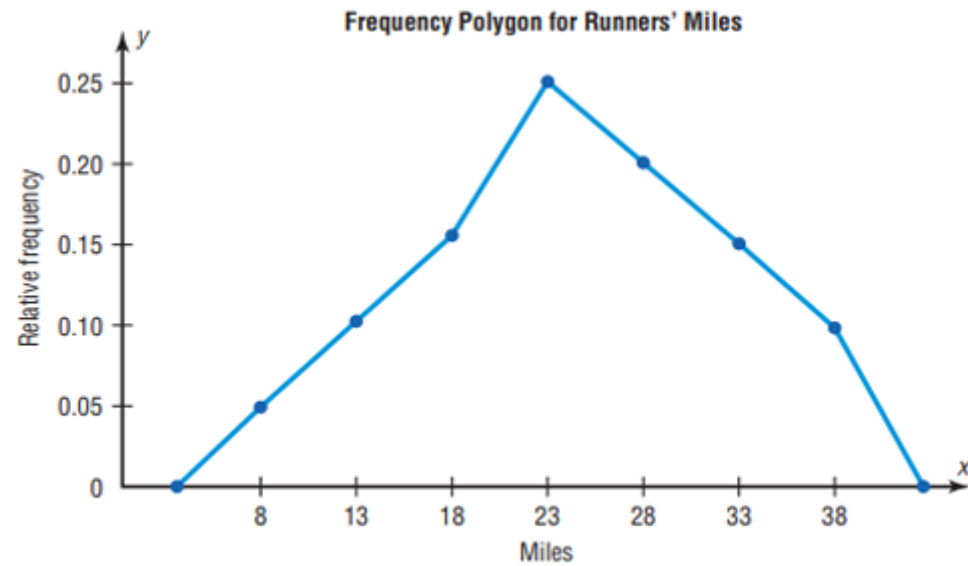
Step 3 Draw each graph as shown in Figure 2–6. For the histogram and ogive, use the class boundaries along the x axis. For the frequency polygon, use the midpoints on the x axis. The scale on the y axis uses proportions.



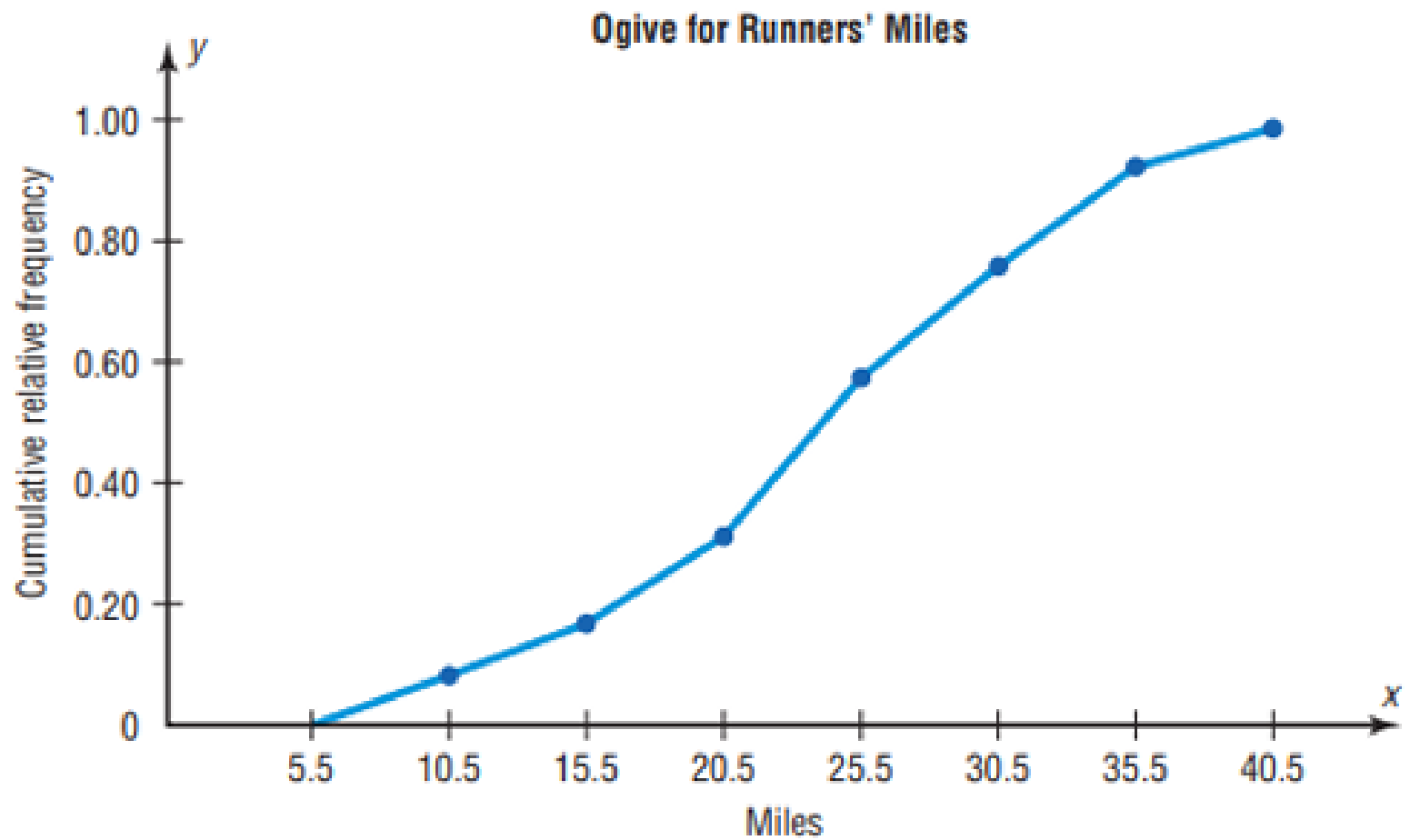
(a) Histogram



(a) Histogram

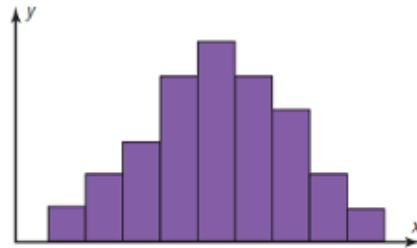


(b) Frequency polygon

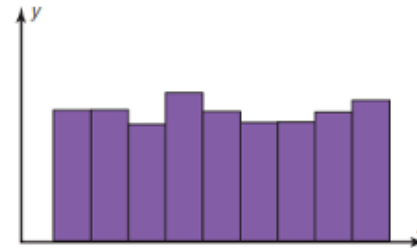


(c) Ogive

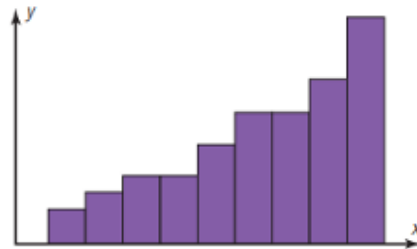
Distribution Shapes



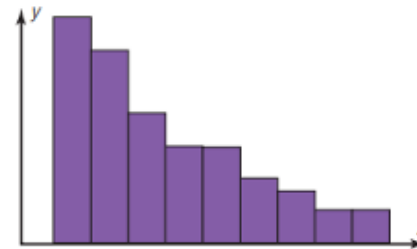
(a) Bell-shaped



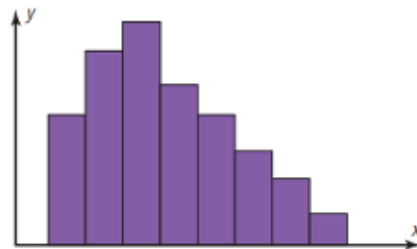
(b) Uniform



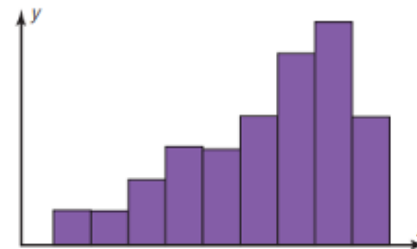
(c) J-shaped



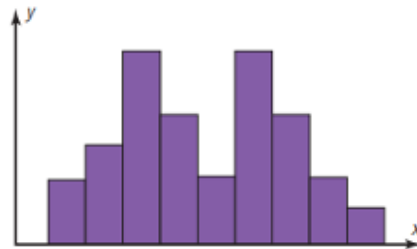
(d) Reverse J-shaped



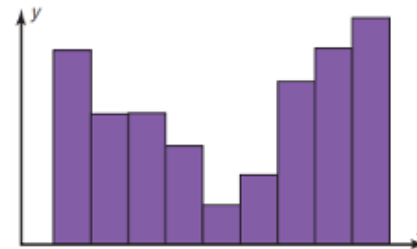
(e) Right-skewed



(f) Left-skewed



(g) Bimodal



(h) U-shaped



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Thank You

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Department of Computer Science & Engineering

